

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Excel-Based Data Analytics & Business Intelligence Project

Prepared by: Ayyam Ullah

Tool Used: Microsoft Excel

Project Type: Customer & Retail Analysis

Executive Summary

This project analyzes customer shopping behavior to identify purchasing patterns, revenue drivers, customer segment differences, and opportunities for business improvement. The analysis was conducted using Microsoft Excel through data validation, analytical column creation, PivotTables, KPIs, charts, and an interactive dashboard.

The analysis examines customer demographics, product categories, purchase amounts, subscription status, and purchasing behavior. The dashboard provides a consolidated view of key performance indicators and revenue patterns, while the supporting analysis converts these findings into actionable business recommendations.

The results indicate that **Clothing is the largest revenue-generating category**, contributing approximately **44.7% of total revenue**, while **Clothing and Accessories together account for approximately 76.5%** of revenue. The analysis also shows differences in total revenue between male and female customers, while average purchase values remain relatively consistent across age groups.

Overall, the project demonstrates how Excel-based data analysis can transform customer transaction data into meaningful business insights that support **inventory planning, marketing strategy, customer segmentation, and revenue growth decisions**.

2. Business Problem & Objectives

2.1 Business Problem

Retail businesses generate large amounts of customer transaction data, but raw data alone does not clearly show **which products generate the most revenue, how customers behave, or which customer segments represent the strongest opportunities**.

The purpose of this project is to analyze customer shopping behavior and transform the available data into meaningful business insights. The analysis focuses on identifying revenue patterns across product categories, demographics, age groups, and subscription status.

The results can help a business make more informed decisions related to **product prioritization, marketing, customer targeting, subscription strategies, and inventory planning**.

2.2 Project Objectives

1. Identify the highest revenue-generating product categories.

2. Compare purchasing behavior between male and female customers.
3. Analyze average purchase values across different age groups.
4. Evaluate purchasing behavior based on subscription status.
5. Identify revenue concentration across products and customer segments.
6. Develop an Excel dashboard to communicate key findings clearly.
7. Provide actionable business recommendations based on the analysis.

2.3 Key Business Questions

- Which product categories generate the most revenue?
- Which categories contribute the least revenue?
- Does purchase value differ significantly across age groups?
- How does revenue differ between male and female customers?
- How does subscription status relate to revenue and average purchase value?
- Which customer and product segments should receive greater business attention?
- What actions could potentially improve revenue and customer engagement?

3. Dataset Overview

3.1 Dataset Description

The dataset contains **3,900 customer shopping records** with **18 variables** covering customer demographics, purchases, products, payment methods, subscriptions, and purchasing behavior.

3.2 Key Data Fields

- Age and Gender
- Product Category
- Purchase Amount
- Subscription Status
- Previous Purchases
- Payment Method
- Review Rating
- Purchase Frequency

3.3 Analytical Columns

Three additional columns were created in Excel to support deeper segmentation:

Analytical Column	Purpose
Age Group	Segments customers into defined age brackets for comparative analysis
Purchase Value Group	Classifies transactions by spending tier
Customer Type	Distinguishes customer categories to support targeted insights

These columns helped segment customers and identify purchasing patterns across the dataset.

4. Data Preparation & Methodology

The dataset was reviewed and prepared in **Microsoft Excel** before analysis. Data quality was checked for missing values and duplicate customer IDs.

Three analytical columns were created to support customer segmentation:

- Age Group
- Purchase Value Group
- Customer Type

The analysis was then performed using **PivotTables, KPIs, charts, and an interactive dashboard** to identify important customer and revenue patterns.

Tools Used

Microsoft Excel | PivotTables | Data Analysis | Data Visualization | Dashboard Design

5. Dashboard Overview

The Excel dashboard provides a concise view of customer shopping performance through **4 key KPIs and 6 visualizations**.

Key Performance Indicators

\$233,081 Total Revenue	\$60.65 Average Purchase	3,900 Total Purchases	150 Subscribers
---	--	---	---------------------------------------

Dashboard Visualizations

The charts analyze revenue and purchasing patterns by:

- Product Category
- Subscription Status
- Age Group
- Gender
- Category & Subscription
- Age Group & Purchase Count

The dashboard allows decision-makers to quickly identify **high-performing categories, customer patterns, and potential growth opportunities.**

6. Key Findings & Insights

The analysis identified the following key findings:

#	Finding
1	Clothing is the leading revenue category, generating \$104,264 (44.7% of total revenue).
2	Accessories are the second-largest category, generating \$74,200 (31.8%).
3	Clothing and Accessories together contribute approximately 76.5% of total revenue.
4	Male customers generate higher total revenue, at \$157,890 compared with \$75,191 from female customers.
5	Average purchase value is relatively consistent across age groups, ranging from \$58.88 to \$60.65.
6	Non-subscribed customers generate higher revenue than subscribed customers in the analyzed records.
7	Average purchase values are similar between subscribed (\$60.13) and non-subscribed customers (\$60.83).

#	Finding
8	Outerwear generates the lowest revenue, at \$18,524 (7.9%).
9	Footwear contributes 15.5% of total revenue, making it a moderate-performing category.
10	Revenue is concentrated in key product and customer segments, highlighting opportunities for focused business strategies.

7. Business Recommendations

Based on the findings, the following actions are recommended:

- Prioritize Clothing and Accessories in inventory and promotional planning.
- Use cross-selling and bundled offers between high-performing categories.
- Develop segment-specific marketing campaigns based on customer purchasing behavior.
- Strengthen the value proposition of subscriptions through rewards, exclusive offers, and discounts.
- Review Outerwear performance and test targeted promotions before increasing inventory.
- Avoid relying heavily on age-based pricing, since purchase values are relatively similar across age groups.
- Investigate the factors behind the gender revenue difference before making major marketing decisions.

8. Conclusion

This project shows how Excel-based analytics can convert raw retail transaction data into clear, decision-ready insights. By combining data preparation, PivotTable analysis, KPI tracking, and dashboard design, the analysis highlights where revenue is concentrated and where the greatest opportunities for growth exist.

Clothing and Accessories stand out as the strongest performing categories and should anchor near-term inventory and marketing decisions, while Outerwear presents an opportunity for targeted review. The consistency of average purchase value across age groups suggests that segmentation strategies should focus more on category and subscription behavior than on age-based pricing alone.